Critical Review for *Transport in Porous Media*

"Development of a Coupled VOF-DEM Solver for Multiphase Fluid Flow in OpenFOAM"

19 July 2026

Recommendation

Major revision.

The manuscript contains a potentially useful solver development and a relevant engineered porous-domain application. It also presents genuine quantitative results, especially the dam-break front curve in Figure 3 and the drainage and retention curves in Figures 7 and 8. However, the current evidence does not establish the paper's main comparative claim that *DPMInterFoam* is more accurate than *MPPICInterFoam* in dense regimes. The study demonstrates a difference between two model formulations, but it does not validate either dense-regime prediction against independent data. Several printed equations also require correction or reconciliation with the code. These issues are substantial but potentially addressable in a major revision.

Summary and positive aspects

The paper develops a VOF-DEM solver in OpenFOAM and compares it with the standard VOF-MPPIC approach. The authors first reproduce a published three-phase dam-break calculation and then study water drainage through a regular array of solid circular obstacles containing mobile particles. The porous-domain study is relevant to the scope of *Transport in Porous Media*, which includes numerical modelling of multiphase transport in engineered porous materials, packed beds, and filtration systems.

Specific strengths are as follows:

- The abstract clearly states the solver, benchmark, application case, comparison, and principal conclusion.
- Figure 2 gives a clear three-dimensional view of the dam-break geometry, while Figure 5 provides the principal dimensions of the porous-domain geometry.
- Figure 3 is a quantitative comparison rather than a purely visual benchmark.
- Figures 7 and 8 report time-resolved injected, drained, and retained water volumes. These are the strongest transport observables in the manuscript.
- The text describes an internal mass-balance cross-check in which retained water is computed both from inlet minus outlet volume and from a spatial integral of the water fraction.
- Table 3 uses the same listed particle count for both solvers in each hydration case. This is an important matched-input feature and should be stated more explicitly.

- The appendices provide useful configuration excerpts. They improve transparency, although they do not replace a complete and versioned code and case archive.

Major comments

1. **The paper demonstrates model discrepancy, not comparative accuracy.**

   The central conclusion is that *DPMInterFoam* is more accurate in dense particle regimes because *MPPICInterFoam* permits particle overlap. The manuscript does show substantial differences in drainage and retention at $\varepsilon_p = 0.1$, 0.2, and 0.3 (Figures 7 and 8). It also proposes a plausible overlap and blockage explanation (Figures 9 and 10). However, neither solver is compared with an experiment or another independently validated dense-bed result for the perforated domain. DEM is therefore being treated as the reference truth without evidence that its contact, drag, porosity-mapping, wetting, and unresolved-coupling choices are accurate in this geometry.

   The defensible conclusion from the present results is narrower: the two closures produce similar bulk drainage at low nominal particle fractions and diverge at higher fractions for the chosen setup. To rank their accuracy, the authors need experimental bed-height, porosity, pressure-drop, particle-motion, or drainage data with uncertainty. Alternatively, the paper should describe a model-form comparison and remove statements that one solver is physically accurate and the other erroneous.

2. **The packing analysis and hydration analysis are not connected clearly enough.**

   On manuscript p. 21, the authors report that filling the lower third of the domain requires 5,606 DEM particles but 94,636 MPPIC particles. Yet Figure 6 shows only the *DPMInterFoam* packing. No matched *MPPICInterFoam* field, bed-height profile, solid-fraction field, overlap distribution, or parcel-weight information is shown. In contrast, Table 3 lists a single common particle count for each hydration case, reaching 10,000 particles at $\varepsilon_p = 0.3$. The manuscript must explain how the 5,606 versus 94,636 packing exercise relates to the later 10,000-particle hydration case.

   For each solver and each case, report:

   - physical particle count, computational parcel count, and particles represented per parcel;
   - initial and time-dependent bed height, total solid volume, and spatial ε_p;
   - an overlap metric, such as normalized interpenetration depth or the fraction of particles exceeding a specified overlap;
   - identical-input and identical-state comparisons as separate tests.

   An identical-input test asks how two closures evolve the same particle inventory. An identical-state test asks how they transport water through matched initial porosity and bed geometry. Both are useful, but they answer different questions.

3. **The governing equations contain substantive notation and force-balance problems.**

   The equation set on manuscript pp. 8 to 10 needs a complete audit against the implemented source code.

   - Equation (1) uses ε_f as a fluid void fraction, but the accompanying text describes it as the volume fractions of the two fluids and separately imposes $\varepsilon_g + \varepsilon_l = 1$. Equation (14) later imposes $\varepsilon_f + \varepsilon_p = 1$. Porosity and phase saturation must be distinguished, for example using ϕ, S_l, and S_g.

- The sentence below Equation (3) states that gas corresponds to $\varepsilon_g = 0$, although the equation is written using the liquid fraction ε_l. Gas should correspond to $\varepsilon_l = 0$ in that definition.
- The gravity term in Equation (2), printed as $\varepsilon_f(\mathbf{g}\nabla\rho_f)$, is not a standard vector body-force term as written. Appendix Figure A.2 shows a p_{rgh} formulation with a density-gradient contribution. The continuous equation, pressure definition, and implemented discretization need to be reconciled explicitly.
- Equation (9) already includes $-V_p\nabla p$, while Equations (11) and (17) include the pressure-gradient force again inside F_{pf}. As printed, the particle pressure force is counted twice.
- In the dilute branch of Equation (14), the hindrance factor is printed as $\varepsilon_p^{-2.65}$. The standard Wen-Yu/Gidaspow form normally uses the fluid fraction in this factor. The authors should verify the printed equation and the exact OpenFOAM implementation.
- Equation (18) applies $V_p\sigma\kappa\nabla\varepsilon_l$ directly as a particle surface-tension force. The manuscript must explain how this term is sampled at a finite-size particle, how wetting and contact angle are handled, and how double counting with the continuum surface-force term is avoided.

These are not merely typographical matters because they determine the momentum exchange being tested.

4. **The dam-break case is a limited benchmark, not full solver validation.**

   Figure 3 shows almost exact overlap of the water-front curves from the reference, *DPMInterFoam*, and *MPPICInterFoam*. This is useful evidence for bulk free-surface propagation. However, no numerical errors, raw data, grid sensitivity, time-step sensitivity, or uncertainty are reported. The remaining column height h^* is defined but not plotted. More importantly, no particle-front, particle-velocity, collision, force, or momentum-exchange observable is compared quantitatively. Figure 4 gives only three qualitative snapshots.

   The claim that the coupled VOF-DEM solver is verified or validated should therefore be limited. At minimum, add:

   - grid and time-step convergence for the water-front and at least one particle observable;
   - liquid, gas, and solid inventory closure;
   - a single-particle settling or drag benchmark;
   - particle-wall and particle-particle collision tests;
   - a two-way momentum-conservation test;
   - direct comparison with the available experiment, not only the previously published numerical curve.

5. **The perforated-domain setup remains under-specified for reproducibility and interpretation.**

   The domain is called two-dimensional on manuscript p. 19, but spherical Lagrangian particles and particle volumes are used. The front-back thickness and the front-back conditions for both fluid and particles must be stated. The term "circular holes" is also misleading because these regions are impermeable solid obstacles in Figure 5 and in the text. They should be called cylinders, posts, or solid obstacles unless they are actual void openings.

   The inlet and outlet velocity, pressure, water-fraction, and particle conditions are not given completely. The paper also needs surface tension, contact angle or wetting treatment, phase initializa-

tion, time-step control, Courant limits, coupling interval, DEM substeps, porosity mapping, force spreading, and fluid solver tolerances.

The mesh is 112×240 over 0.14×0.30 m, giving a nominal cell size of about 1.25 mm. The particle diameter is 1 mm, and the minimum gap between adjacent 25.4 mm obstacles spaced 31 mm apart is about 5.6 mm. Thus, the narrowest gap is resolved by only about 4.5 nominal cells, and the cell size is comparable to the particle diameter. Mesh sensitivity and porosity-mapping sensitivity are essential for the reported drainage and blockage behavior.

6. **Figures 7 and 8 are valuable, but the simulations are too short to support "ultimate" drainage claims.**

   The drainage and retained-volume curves are a strong part of the paper. Nevertheless, several dense-case curves, especially for *MPPICInterFoam*, are still evolving at the 6 s endpoint. The paper therefore cannot conclude that one solver retains water "by the end" in a physically asymptotic sense or that the other cases reach the same ultimate drained volume.

   Extend each run until a predefined drainage-rate or retained-volume convergence criterion is met. Show the mass-balance residual explicitly rather than only stating that two calculations agree. Report the two retained-volume estimates and their difference over time. Because particle injection and initial packing can be stochastic, repeat the cases or demonstrate that the initial states and results are deterministic. Add variability bands where appropriate.

7. **The proposed overlap mechanism is not measured in the figures used to support it.**

   Figure 9 compares water volume fraction for both solvers over five particle fractions and four times, which is useful. However, it does not clearly display particle centers, particle overlap, or a quantitative porosity field. Figure 10 enlarges the water field and labels regions as blocked by overlap, but the overlap itself is not quantified and the grey particle region is not explained by a legend. The figure therefore illustrates a hypothesis rather than demonstrating its cause.

   Add matched panels of water fraction, particle locations, solid fraction, and a direct overlap or porosity diagnostic. A profile of ε_p and liquid flux across the lower rows, together with pressure drop and particle export, would connect the proposed mechanism to the drainage curves.

8. **Computational-cost conclusions need correction and uncertainty.**

   Table 3 is useful because it reports absolute wall times at matched listed particle counts. However, the first percentage is incorrect: $(930-928)/928 \times 100\% = 0.216\%$, not 0.02%. Timing is based on a single run per case, with no variability, step count, nonlinear iteration count, adaptive time-step history, or memory use. The paper also compares serial timing for the dam break with 20-core timing for the porous case. Report repeated timings, compiler and build type, OpenFOAM version, actual core use, and solver iteration statistics. The current timing results should be described as case-specific measurements, not general efficiency limits.

9. **The open-source claim must be made reproducible.**

   The manuscript repeatedly presents *DPMInterFoam* as an open-source contribution and links to a GitHub repository on p. 7. As checked on 15 July 2026, that public repository exists but is empty.[1] Appendix screenshots are not sufficient to compile or reproduce the solver. A revision should provide a tagged release with a license, source code, OpenFOAM dependency and version, build instructions, complete case directories, post-processing scripts, and source data for every plotted curve and field.

[1] https://github.com/iMSEL-USC/DPMInterFOAM, accessed 15 July 2026.

10. **The abstract and conclusion should be narrowed to the evidence.**

    The abstract is complete and appropriately concise. Its final claims are nevertheless too broad. The paper has not shown that *DPMInterFoam* models real-world multiphase systems accurately, nor that it is more accurate across dense regimes generally. The conclusion on manuscript p. 29 extends the solver recommendation to bubbling, circulating flow, gasification, carbon capture, and fluidized-bed deposition, none of which is simulated here. These applications should be removed or presented explicitly as future tests. The conclusion should state the demonstrated result: agreement in the selected water-front benchmark and model-dependent divergence in a particular porous drainage setup.

Minor and editorial comments

- The manuscript uses eight keywords, while the current journal instructions request four to six. Three short Article Highlights are also requested. The abstract itself already satisfies the stated length requirement.
- On p. 22, the dense case is introduced once as $\varepsilon_p \approx 0.22$, whereas Figures 7 to 9 and Table 3 use 0.2. Use one value consistently and explain how it is calculated.
- Replace "Nomenclatures" and "Abbreviation" with "Nomenclature" and "Abbreviations." Distinguish porosity, phase fraction, and particle volume fraction in the symbol list.
- Replace "hydration analysis" with "drainage and retention analysis" unless hydration has a specific constitutive meaning in this study.
- References [1]/[24], [3]/[10], [9]/[36], and [19]/[26] are duplicates. Reference [28] is incomplete. Reformat the references in the journal's author-year style and verify all bibliographic fields and DOI links.
- Remove the yellow text highlighting visible around Figure 9 cross-references on manuscript p. 24.
- Replace claims such as "ground truth," "physically accurate," and "real-world ... accurately" with language matched to the actual evidence.

Final assessment

The manuscript is not ready for acceptance, but it contains more substantive evidence than a simple solver demonstration. In particular, Figures 7 and 8 provide a useful transport dataset, and Table 3 establishes a matched particle-count comparison for the hydration cases. The paper's main weakness is the interpretation of solver disagreement as proof of comparative accuracy. If the authors correct the equations, provide reproducible code, quantify the proposed overlap mechanism, and validate or appropriately narrow the dense-regime claim, the work could become a credible contribution to numerical multiphase transport in engineered porous media.